

Richard Motorgeanu

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Highlights of Qualifications

- Enrolled in Level 3 of the Honours Computer Science program at McMaster University
- Extensive coding experience in OOP languages, such as C#, C++, and Python, established by course work and personal projects
- Familiar with version control systems, in particular Git, as shown through course work and personal projects

Education

Bachelor of Applied Science | Honours Computer Science (Co-op) Sep 2022 - Apr 2027 (projected)
McMaster University, Hamilton ON GPA: 4.00

- Relevant courses: DSA, Computer Architecture, Computer Graphics, Physics for Video Games
- Fields of interest: Game Programming, Engine Development, Software Development and Design

Work Experience

Learning Resource Technical Developer Sep 2024 - Present (part-time)
McMaster University, Hamilton ON May 2024 - Aug 2024 (co-op)

- Started to co-design and develop a README generator tool that would help researchers create READMEs for their datasets with ease. Prototyped in HTML, CSS, JS, to be moved to Drupal over the Fall
- Created a program that would query the Echo360 API, filter out videos created by the centre, and collect/aggregate relevant statistics and data for over 200 workshop videos automatically using Python and GitHub Actions
- Maintained, updated, and archived old repositories, websites, videos, and documentation

Learning Resource Development Assistant Sep 2023 - Apr 2024 (part-time)
McMaster University, Hamilton ON May 2023 - Aug 2023 (co-op)

- Collaborated with Sherman Centre staff to develop material created from previous workshops into asynchronous learning modules that meet the needs of learners within and beyond McMaster University
- Used pedagogical best practices to create highly engaging and effective learning resources that may be reused by instructors and learners in face to face, hybrid, or virtual learning environments
- Utilized GitHub Pages and Jekyll to host numerous modules, customizing sites using Sass to align with the branding standards of the Sherman Centre and McMaster University, and ensuring compliance with accessibility guidelines through the use of WAVE

Student Leader | Summer Coding Camp Jul 2022
Peel District School Board, Mississauga ON

- Became familiar with tools and languages to support the summer camp students, encouraging independent learning through the use of tutorials and experimentation
- Planned and created lesson plans and project outlines to assist student's learning and creativity, resulting in over 30 unique projects being created as part of their final inquiry project
- Aided students individually and in small groups with coding errors, equipping students with effective problem solving skills

Projects

Linear Lab | Class Project

Feb 2023 - May 2023

- Led a team of 4 skilled individuals in designing and developing an interactive textbook tool to help students learn about and visualize complex linear algebra concepts
- Applied design thinking by iterating through weekly prototypes and conducting user feedback interviews to ensure continuous improvement and alignment with user needs

Nyctomorph | Game Jam Project

Feb 2023

- Collaborated with a friend to develop a 2D survival game using Godot 3 in 9 days for the 5th NOKIA 3310 game jam
- Designed the game under strict visual requirements, writing shaders to restrict the color palette

Crop Rush | Game Jam Project

Jan 2023

- Developed a 2D puzzle game using the Unity Game Engine in 72 hours for the Ludum Dare 52 game jam
- Gathered feedback from players and implemented new features to meet player expectations
- Scored 226th out of 825 entries in the Fun category, and 358th overall

KTaNE Voice Assistant Bot | Personal Project

Jan 2022 - Oct 2022

- Created an interactive voice assistant bot in Python that acts as the second player for the popular party game *Keep Talking and Nobody Explodes*
- Applied important software design principles and best practices, such as modularity
- Provided documentation for the code, as well as an installation and usage guide using GitHub Wikis in order to assist future contributors and users and promote sustainable development

Volunteer Work

Research Assistant

Feb 2023 - Apr 2023

The Gaming Scalability Environment (G-Scale) Lab, Hamilton ON

- Researched game design and how certain design choices affect cognitive and motor requirements
- Designed 3 mini-games that test a user's type/token change detection, time to contact, and heading/steering skills
- Developed a prototype baseball mini-game that tracks and records data regarding a user's time to contact skills

Extracurricular Activities

Varsity Fencing

Sep 2022 - Present

McMaster University, Hamilton ON

- Demonstrating time management by actively commuting to attend biweekly practices and lessons while maintaining school work and long commuting schedule
- Participating in solo and team tournaments on a monthly basis
- Leading warm-up exercises and demonstrating new skills and techniques to beginner fencers

Skills

Languages:

Python, C#, C++, C, Java, HTML/JS/CSS, SASS, $\text{\LaTeX}$, SQL, Haskell, R

Tools and Frameworks:

OpenGL, Unity, Godot, Git, GitHub, GitHub Pages, Jekyll, VSCode, Visual Studio, Microsoft 365, AutoCAD

Spoken Languages:

English, French, Romanian